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Comparative Analysis of Lipid Profiles Between Controlled and Uncontrolled Type 2 Diabetes Mellitus Patients at Tarakan Regional General Hospital (2022–2023)

ABSTRACT

Background: Patients with type 2 diabetes mellitus (T2DM) often experience abnormalities in lipid metabolism, which contribute to an increased likelihood of cardiovascular complications, especially in the presence of poor glycemic control. Despite this, evidence derived from hospital-based populations in Indonesia is still limited.

Methods: A cross-sectional analysis was conducted using medical record data from 120 patients diagnosed with T2DM. Glycemic control was determined based on HbA1c values and categorized into controlled ($<7\%$) and uncontrolled ($\geq 7\%$) groups. The lipid profile parameters evaluated included total cholesterol, LDL-C, HDL-C, and triglycerides. Statistical analysis was performed using both parametric and non-parametric methods as appropriate.

Results: Individuals with uncontrolled glycemic status tended to exhibit higher concentrations of total cholesterol, LDL-C, and triglycerides compared to those with controlled diabetes. Meanwhile, HDL-C levels showed no significant difference between the two groups. A higher prevalence of dyslipidemia was also observed among patients with inadequate glycemic control.

Conclusion: Poor glycemic control is associated with adverse changes in lipid profiles among patients with T2DM, highlighting the need for comprehensive management approaches that address both glycemic status and cardiovascular risk factors.

Keywords: diabetes mellitus, HDL, total cholesterol, LDL, triglycerides

12 INTRODUCTION

The increasing prevalence of type 2 diabetes mellitus (T2DM) has become a major concern due to its contribution to long-term vascular complications, particularly cardiovascular disease. Indonesia is among the countries with a substantial number of diabetes cases, with approximately 19.5 million individuals affected in 2021.¹ National surveys have also demonstrated a steady rise in prevalence from 6.9% in 2013 to 8.5% in 2018, indicating a growing public health burden.²

Metabolic dysregulation in T2DM²⁸ is closely linked to insulin resistance and chronic hyperglycemia, which⁶ interfere with normal lipid homeostasis. These disturbances often result in an unfavorable lipid profile characterized by elevated triglycerides and low-density lipoprotein cholesterol (LDL-C), along with reduced high-density lipoprotein cholesterol (HDL-C).³ Such alterations contribute to the progression of atherosclerosis and increase the likelihood of cardiovascular complications.^{3–5} Prolonged exposure to hyperglycemia may also trigger oxidative stress and endothelial dysfunction, further accelerating vascular damage.^{3–5}

Glycemic status, commonly assessed using glycated hemoglobin (HbA1c), is an important determinant of these metabolic changes. Patients with poor glycemic control tend to exhibit more pronounced lipid abnormalities compared with those whose glycemic levels are adequately controlled.^{4,5} Despite this well-established relationship, evidence derived from hospital-based populations in Indonesia remains scarce.

Based on this background, the present study was designed to evaluate differences in lipid profiles between controlled and uncontrolled T2DM patients and to explore their clinical relevance in relation to cardiovascular risk.

METHODS

This research used an analytic cross-sectional design²⁹ and was conducted at a tertiary care hospital in Jakarta, Indonesia, during October 2024. Ethical clearance for the study was granted by the Institutional Ethics Committee (No. 1811/SLKE/TM/UKK W/FKIK/KEPK/TX/2024).

A total of 120 patients diagnosed with type 2 diabetes mellitus (T2DM) were selected from medical records using a consecutive sampling technique. The diagnosis of T2DM was based on documented glycated hemoglobin (HbA1c) levels $\geq 6.5\%$ or fasting plasma glucose ≥ 126 mg/dL. Patients were excluded if they had incomplete lipid profile data, were undergoing lipid-lowering treatment (such as statins, fibrates, niacin, or ezetimibe), or had comorbid conditions including chronic kidney disease, hyperthyroidism, or liver cirrhosis.

Participants were grouped according to glycemic status into controlled (HbA1c $< 7\%$) and uncontrolled (HbA1c $\geq 7\%$) categories. In this study, glycemic control functioned as the independent variable, while lipid profile indicators—namely triglycerides, LDL-C, HDL-C, and total cholesterol—were designated as dependent variables. Lipid profile data were obtained from medical records and evaluated within an appropriate timeframe in relation to glycemic assessment.

Statistical analysis was performed using SPSS version 24. Data distribution was examined prior to analysis. For variables with normal distribution, comparisons were conducted using the independent t-test, whereas the Mann–Whitney U test was applied for non-normally distributed data. Statistical significance was defined as $p < 0.05$.

RESULTS

A total of 120 patients with type 2 diabetes mellitus (T2DM) were analyzed. The largest proportion of participants was in the 45–59 year age group, and female patients constituted the majority. Based on body mass index classification, Obesity Class I was the most frequently identified category. A higher portion of patients were classified as having uncontrolled glycemic status.

Table 1 presents the demographic and clinical characteristics of the study population.

Table 1. Distribution of Study Participants According to Demographic and Clinical Characteristics

Characteristics	Frequency (%)	Cholesterol Level			
		Total Cholesterol	Triglycerides	LDL	HDL
(mg/dL)					
Age(years)					
Adult(19-44)	10 (8,3)	182,5 (105-324)	130 (68-193)	119 (60-225)	42 (12-60)
Pre-Elderly(45-59)	62 (52,7)	212 (108-323)	160 (57-828)	121 (46-211)	46 (23-75)
Elderly (≥60)	48 (40,0)	191 (95-299)	131 (49-366)	106 (19-209)	45 (23-79)
Sex					
Male	47 (39,2)	177 (95-315)	142 (49-828)	110 (19-209)	44 (12-73)
Female	73 (60,8)	216 (123-343)	143 (57-617)	129 (45-225)	50 (23-79)
BMI (kg/m²)					
Underweight (<18,5)	7 (5,8)	205 (145-218)	195 (75-285)	121 (80-144)	43 (26-60)
Normal (18,5-22,9)	14 (11,7)	182 (135-235)	116 (57-213)	101 (45-159)	43 (27-79)
Overweight(23-24,9)	24 (20,0)	186 (108-343)	129 (49-480)	108 (46-209)	45 (23-75)
Obesity Class I (25-29,9)	49 (40,8)	212 (95-324)	149 (84-828)	124 (19-225)	47 (12-75)
Obesity Class II (≥30)	26 (21,7)	194 (123-264)	150 (66-385)	117 (75-211)	47 (23-74)
Glycemic Control Status					
Controlled (< 7%)	43 (35,8)	179,23 ±43,24	128 (49-385)	109,70 ±38,66	45,86 ±13,66
Uncontrolled (≥ 7%)	77 (64,2)	210,86 ±51,64	156 (57-828)	122,62 ±37,72	47,29 ±13,68

Variations in lipid values were observed across demographic groups. Higher lipid levels were generally identified among individuals aged 45–59 years, female patients, and those categorized as Obesity Class I.

The distribution of dyslipidemia according to glycemic status is shown in Table 2.

Table 2. Distribution of Dyslipidemia Among Patients with Type 2 Diabetes Mellitus

Dyslipidemia	Glycemic Control Status	
	Controlled	Uncontrolled

	(n=43)	(n=77)
	n (%)	
Dyslipidemia	4 (3,3)	12 (10,0)

A greater proportion of dyslipidemia was observed among patients with uncontrolled glycemic status compared with those who were controlled.

Comparison of lipid parameters between controlled and uncontrolled T2DM patients is presented in Table 3.

Table 3. Comparison of Lipid Parameters According to Glycemic Control Status

Lipid Profile	Glycemic Control Status		Nilai p
	Controlled (n=43)	Uncontrolled (n=77)	
LDL	109,70 ±38,66	122,62±37,72	0,047 ^a
HDL	45,86±13,66	47,29±13,68	0,585 ^a
Kolesterol Total	179,23±43,24	210,86±51,64	0,001 ^a
Trigliserida	131,66 (49-285)	156,00 (57-828)	0,013 ^b

^aUji T test independent, ^bUji mann whitney, significant if p<0,05

Higher concentrations of total cholesterol, LDL-C, and triglycerides were observed among patients with poor glycemic control relative to those with controlled diabetes. In contrast, no significant variation in HDL-C levels was identified between the groups.

DISCUSSION

The age distribution observed in this study indicates that T2DM predominantly affects individuals within the 45–59 year range. This pattern is consistent with previous evidence suggesting that metabolic disorders become more prominent with advancing age.^{9,10} Age-related physiological changes, combined with lifestyle factors, may contribute to impaired regulation of glucose and lipid metabolism.^{10,11}

A higher proportion of female patients was identified, and this group tended to demonstrate less favorable lipid profiles. Previous studies have also reported similar trends, indicating a greater susceptibility to dyslipidemia among women with T2DM.^{7,12} Hormonal changes, particularly declining estrogen levels, may influence fat distribution and insulin sensitivity, thereby affecting lipid metabolism.¹¹

The majority of patients were categorized as Obesity Class I, with this group exhibiting elevated total cholesterol and LDL-C levels. These findings support existing evidence linking obesity with lipid abnormalities in T2DM.^{13,14} Increased adipose tissue contributes to higher levels of circulating free fatty acids and inflammatory mediators, both of which may interfere with insulin signaling and promote hepatic lipoprotein production.^{13,15}

Lipid abnormalities were also identified in individuals with lower body mass index, particularly in the form of elevated triglyceride levels. Although less frequently discussed, this observation suggests that dyslipidemia in T2DM may arise from multiple metabolic pathways beyond obesity alone.^{14,16} This reinforces the complexity of lipid metabolism disturbances in diabetic conditions.

Patients with uncontrolled glycemic status demonstrated ¹ higher levels of total cholesterol, LDL-C, and triglycerides compared with those who achieved glycemic control. This pattern aligns with previous research indicating that poor glycemic regulation is associated with a more atherogenic lipid profile.^{4,5,15} Elevated LDL-C levels may be related to increased hepatic production of very low-density lipoprotein (VLDL), which subsequently contributes to the formation of atherogenic particles.^{18,19}

¹⁴ No significant difference in HDL-C levels was observed between the two groups. Although earlier studies generally report reduced HDL-C levels in T2DM, variations in clinical characteristics, adherence to therapy, or sample size may explain this finding.¹⁹

Elevated triglyceride levels among patients with poor glycemic control further highlight the role of insulin resistance in lipid metabolism. Increased lipolysis leads to a greater release of free fatty acids, which enhances hepatic triglyceride synthesis and contributes to dyslipidemia.¹⁹

Taken together, these findings underscore ²¹ the importance of maintaining adequate glycemic control in patients with T2DM. Poorly controlled diabetes is associated with more unfavorable lipid profiles, which may increase cardiovascular risk. Therefore, comprehensive management strategies that address both glycemic status and lipid abnormalities are essential for improving patient outcomes.²⁰

CONCLUSION

These results ²⁷ highlight the important role of glycemic control in determining ²² lipid profile alterations among patients with type 2 diabetes mellitus. Patients with uncontrolled diabetes exhibited higher concentrations of total cholesterol, LDL-C, and triglycerides, indicating a more atherogenic pattern and a higher likelihood of cardiovascular risk. Notably, lipid disturbances were observed across different BMI categories, including underweight individuals, indicating that dyslipidemia in T2DM is not solely attributable to obesity but arises from complex metabolic interactions involving insulin resistance, glycemic control, and lipid metabolism. ⁴ These findings underscore the importance of comprehensive metabolic assessment and strict glycemic management, alongside routine lipid monitoring, to reduce cardiometabolic risk and improve long-term outcomes in patients with T2DM.

² Conflict of Interest

The authors declare that there are no conflicts of interest regarding the publication of this study.

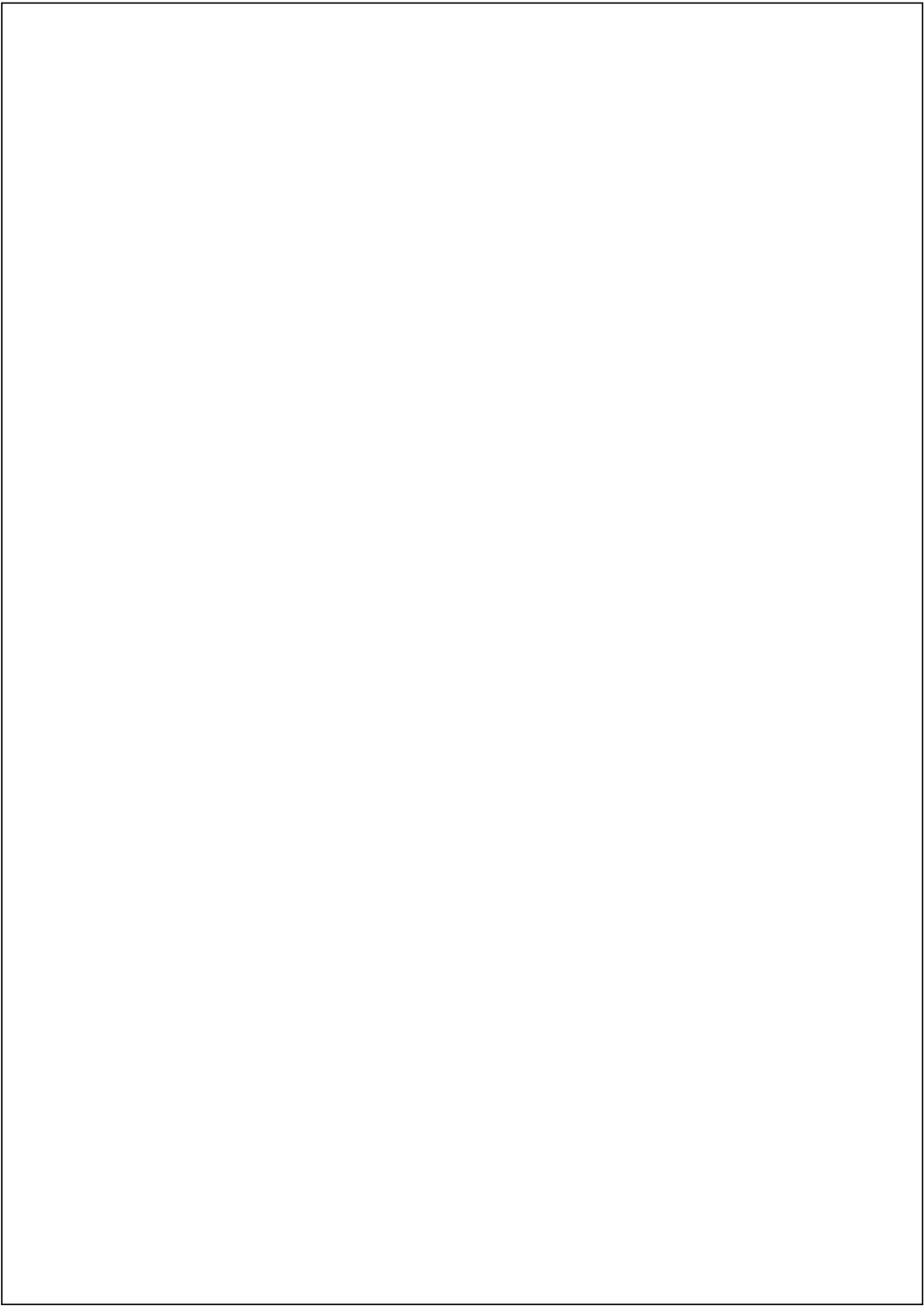
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